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Chromosome-level genome assembly of the obligate cave loach *Troglonectes daqikongensis*

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Troglonectes daqikongensis **is an obligate cave-dwelling fish with a narrowly restricted distribution in southwestern China and pronounced troglomorphic traits. Despite its ecological and evolutionary significance, genomic resources for this species remain scarce. Here, we report a chromosome-level reference genome of *Troglonectes daqikongensis*, assembled using PacBio HiFi long reads, Hi-C data, and MGI T7 short reads. The final assembly spans 615.17 Mb across 47 scaffolds, with a contig N50 of 20.92 Mb and a scaffold N50 of 23.57 Mb. In total, 97.96% of the assembly was anchored to 25 pseudo-chromosomes. Repetitive elements comprise 276.51 Mb, accounting for 44.95% of the genome. A total of 24,652 protein-coding genes were predicted using a combination of *de novo* prediction, homology-based annotation, and transcriptome evidence, with 99.89% functionally annotated. BUSCO assessments indicated high completeness for both the genome assembly (96.6%) and gene annotation (95.5%). This high-quality genome provides a foundational resource for evolutionary and ecological research, including studies on the genetic basis of cave adaptation and the conservation of the microendemic species *Troglonectes daqikongensis*.**

Background & Summary

The blind cave loach, *Troglonectes daqikongensis*, is an obligate troglobitic species in the genus *Troglonectes* (family Nemacheilidae, order Cypriniformes). Initially described as a member of the genus *Oreonectes* in 2016, it was reassigned to *Troglonectes* in 2019 based on morphological and phylogenetic evidence[1,2]. As a typical cave-adapted fish, *Troglonectes daqikongensis* exhibits hallmark troglomorphic traits, including complete eye degeneration, loss of pigmentation, absence of scales, and prominent fat deposition in the caudal peduncle and abdominal cavity. The body is semi-transparent and white, with internal organs visible from the ventral side, and melanin production is absent even under light exposure. The species is endemic to the Daqikong Cave and nearby karst systems in Libo County, Guizhou, China, where it inhabits aphotic subterranean waters and is often the only fish species present[1].

Although currently listed as Least Concern by the International Union for Conservation of Nature (IUCN) Red List of Threatened Species[3], *Troglonectes daqikongensis* has a narrow distribution and strict ecological dependence on cave habitats, which may be affected by anthropogenic factors such as groundwater contamination, tourism development, and biological invasions[4–6]. Further genomic and ecological investigations could support a better understanding of its habitat specificity and potential vulnerability to environmental change[5].

Interest in the adaptive evolution of cave-dwelling organisms dates back to Darwin, who, in *On the Origin of Species*, speculated on the degeneration of eyes in cave fauna: "As it is difficult to imagine that eyes, though useless, could be in any way injurious to animals living in darkness, their loss may be attributed to disuse"[7]. Extreme cave environments, characterized by perpetual darkness and limited availability of food and other resources, impose strong selective pressures that drive the evolution of distinct morphological, behavioral, and physiological traits, often divergent from those of surface-dwelling relatives[4,8–10]. Among these, eye degeneration

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and pigment loss are the most prominent and widespread troglomorphic adaptations[11,12]. In addition to their evolutionary significance, cavefish have also emerged as model systems for biomedical research. Their unique adaptations, such as insulin resistance and ocular degeneration, provide insights into human diseases, including diabetes and ocular disorders[5,13–15].

Although the Mexican tetra (*Astyanax mexicanus*) has long served as the principal model for studying the genetic basis of cave adaptation[8,9], lineage-specific evolutionary trajectories may differ considerably. Expanding genomic investigations to a broader array of cave-adapted lineages is therefore essential for uncovering general and lineage-specific mechanisms of subterranean adaptation[16,17].

The extensive karst regions of southwestern China—characterized by limestone caves and subterranean rivers—harbor more than 150 endemic cavefish species[4,18]. However, existing genomic studies have largely focused on representatives of the genera *Triplophysa* (family Nemacheilidae) and *Sinocyclocheilus* (family Cyprinidae)[16,17], leaving a gap in genomic resources for obligate cave-dwelling species of *Troglonectes*.

With its pronounced troglomorphic features—visual degeneration, depigmentation, fat accumulation, and likely physiological modifications—*Troglonectes daqikongensis* serves as a model for investigating the genetic underpinnings of cave adaptation in fishes. Nevertheless, the lack of genomic resources has limited exploration of its evolutionary history and molecular adaptations. Moreover, assembling a high-quality genome is essential for evaluating genetic diversity and developing evidence-based conservation strategies for this narrowly endemic species[19].

To fill this gap, we assembled a chromosome-level genome of *Troglonectes daqikongensis* using PacBio HiFi long-read and Hi-C sequencing data. The final assembly spans 615.17 Mb and comprises 25 pseudo-chromosomes and 22 unanchored scaffolds, with high continuity (contig N50 = 20.92 Mb; scaffold N50 = 23.57 Mb) and completeness, as validated by BUSCO and other quality metrics. This high-quality genomic resource enables future studies on the genetic basis of cave adaptation, phenotypic evolution, and conservation of this microendemic species.

Methods

Sample collection and sequencing. An adult female *Troglonectes daqikongensis* (Fig. 1) was collected from a subterranean river in the Dagou River system, located in Daqikong Scenic Area, Libo County, Guangxi Province, China, near the species' type locality. The specimen was anesthetized using MS-222 prior to tissue sampling[20]. Nine tissue types, including muscle, brain, gill, kidney, liver, olfactory organ, ovary, skin and swim bladder were dissected, immediately flash-frozen in liquid nitrogen, and stored at −80 °C until further use.

High-molecular-weight genomic DNA was extracted from muscle tissue using the MagAttract HMW DNA Kit (Qiagen, USA) for genome sequencing. DNA integrity was assessed by 1% agarose gel electrophoresis, while purity and concentration were measured using a NanoDrop One spectrophotometer (Thermo Fisher Scientific, USA) and a Qubit Fluorometer (Invitrogen, Thermo Fisher Scientific, USA), respectively.

A short-read library with an average insert size of 350 bp was prepared using the MGIEasy Universal DNA Library Prep Set (MGI, China), and sequenced on the DNBSEQ-T7 platform (MGI, China) using a paired-end 150 bp (PE150) strategy. This generated a total of 32.03 Gb of raw paired-end reads, providing approximately 52.07× coverage of the genome (Table 1). All subsequent calculations of coverage depth in this study were estimated based on the final assembled genome size of 615.17 Mb.

For PacBio HiFi long-read sequencing, libraries were prepared using the SMRTbell Template Prep Kit 2.0 (PacBio, CA, USA) following the manufacturer's standard protocol. The library was sequenced on 0.5 SMRT cells on the PacBio Revio system in Circular Consensus Sequencing (CCS) mode, following PacBio's recommended multiplexing practice. Specifically, our library was pooled with a phylogenetically distant species in the same SMRT cell, and the data were demultiplexed based on their unique library barcodes. This run yielded 41.18 Gb of HiFi reads with an N50 read length of 17.69 kb, corresponding to approximately 66.93 × genome coverage (Table 1).

For Hi-C sequencing, Hi-C libraries were prepared from the muscle tissue of the same specimen using an in-house protocol based on a standard Hi-C methodology. In brief, chromatin was cross-linked with formaldehyde, and nuclei were extracted and purified through a series of sucrose cushion centrifugations. The chromatin was then digested with the MboI restriction enzyme (NEB). Following digestion, DNA ends were labeled with biotin-14-dCTP and subjected to proximity ligation. After reversing cross-links and extracting DNA, biotin-labeled ligation junctions were captured and enriched using streptavidin magnetic beads (Invitrogen). The final libraries were constructed through sonication, end repair, adapter ligation, and PCR amplification. These libraries were sequenced on the DNBSEQ-T7 platform (MGI, China) with a paired-end 150 bp (PE150) strategy. A total of 63.95 Gb of Hi-C data was generated, providing approximately 103.95 × genome coverage for scaffolding and chromosomal assembly (Table 1).

For transcriptome sequencing, total RNA was extracted separately from nine tissues (brain, gill, kidney, liver, muscle, olfactory organ, ovary, skin, and swim bladder) using TRIzol reagent (Invitrogen, Carlsbad, CA, USA). RNA concentration was measured with a Qubit Fluorometer, and RNA quality was evaluated using a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific, USA) and an Agilent 2100 Bioanalyzer (Agilent Technologies, Palo Alto, CA, USA). cDNA libraries were constructed from each RNA sample using the MGIEasy Universal DNA Library Prep Set (MGI, Shenzhen, China), and sequenced on the DNBSEQ-T7 platform (MGI, China) with a paired-end 150 bp (PE150) strategy. In total, 73.77 Gb of raw transcriptomic data were generated across all tissues (Table 1).

Genome Survey and Mitochondrial Genome Assembly. A genome survey was conducted using short-read NGS data from DNBSEQ-T7 to characterize the genomic features of *Troglonectes daqikongensis*.

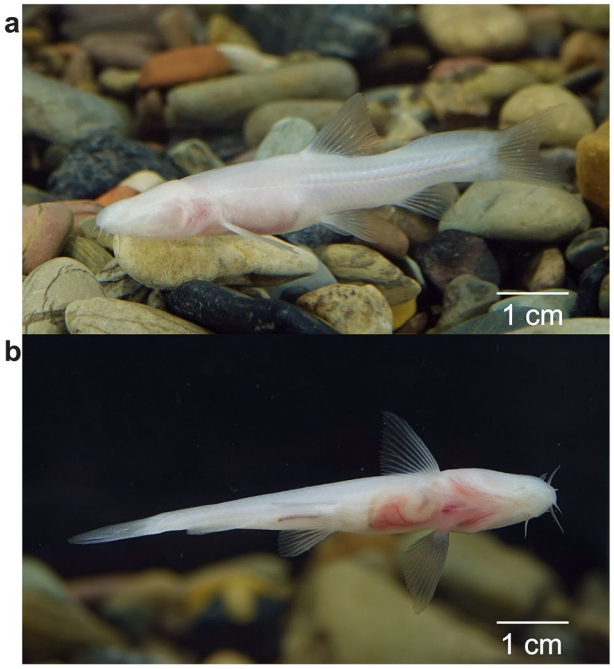


Fig. 1 Specimen of *Troglonectes daqikongensis* used for genome sequencing and assembly. (**a**) Lateral view. (**b**) Ventral view.

Library types	Sample	Bases (bp)	Reads/Pairs	Mean Reads length (bp)	N50 (bp)	Coverage Depth (×)
Short-read NGS	Muscle	32034148200	106780494 pairs	150	150	52.07
Pacbio HiFi	Muscle	41175967077	2408518 reads	17096	17694	66.93
Hi-C	Muscle	63948295800	213160986 pairs	150	150	103.95
RNA-seq	Brain	6684814500	22282715 pairs	150	150	—
RNA-seq	Gill	7863714600	26212382 pairs	150	150	—
RNA-seq	Kidney	13544772600	45149242 pairs	150	150	—
RNA-seq	Liver	6706739100	22355797 pairs	150	150	—
RNA-seq	Muscle	8901540000	29671800 pairs	150	150	—
RNA-seq	Olfactory organ	6188615100	20628717 pairs	150	150	—
RNA-seq	Ovary	7869381600	26231272 pairs	150	150	—
RNA-seq	Skin	8591220000	28637400 pairs	150	150	—
RNA-seq	Swim bladder	7418317200	24727724 pairs	150	150	—

Table 1. Summary statistics of sequencing data used for genome assembly and annotation of *Troglonectes daqikongensis*. Notes:Coverage depth was estimated based on the final assembled genome size (615.17 Mb).

Raw reads were quality-filtered and adapter-trimmed using fastp (v0.23.4; parameters: -f 15)[21]. A 21-mer frequency distribution was generated with Jellyfish (v2.2.10; parameters: -m 21 -C -s 1 G–min-quality = 20)[22], and genome size estimation was performed using GenomeScope2.0 (v2.0.1; parameters: -k 21)[23]. The genome of *Troglonectes daqikongensis* was estimated to be approximately 575.36 Mb in size, with a heterozygosity rate of 0.176% and repeat content of 27.9% (Fig. 2).The filtered short reads were then assembled into scaffolds using SPAdes (v4.0.0)[24]. Assembly quality was assessed with QUAST (v5.1.0)[25], which indicated a draft genome size of 567.07 Mb (Table 2).

To validate species identification, we first performed *de novo* assembly of the mitochondrial genome using second-generation short-read NGS data with GetOrganelle (v1.7.5; parameters: -k 21,45,65,85,105 -F animal_mt -R 10)[26], resulting in a complete circular mitochondrial genome of 16579 bp. Annotation with MitoFish (https://mitofish.aori.u-tokyo.ac.jp/)[27] identified 13 protein-coding genes, 22 tRNA genes, and 2 rRNA genes. The *cytb* gene sequence was aligned against the NCBI Nucleotide (Nt) database using BLASTn (https://blast.ncbi.nlm.nih.gov/Blast.cgi). Validation based on nucleotide identity confirmed the species identity as *Troglonectes daqikongensis*, showing 100.00% sequence identity with four NCBI reference records (the complete mitochondrial genomes MG738689.1 and NC_073146; and the cytb genes OK001253.1 and KU987436.1).

To obtain a more accurate mitochondrial genome assembly and validate the reliability of the short-read (NGS) based assembly, we performed independent assembly of PacBio HiFi data using MitoHiFi (v3.2.1; parameters: -g NC_073146.gbk -f NC_073146.fa -a animal)[28], followed by annotation via MitoFish. This produced a

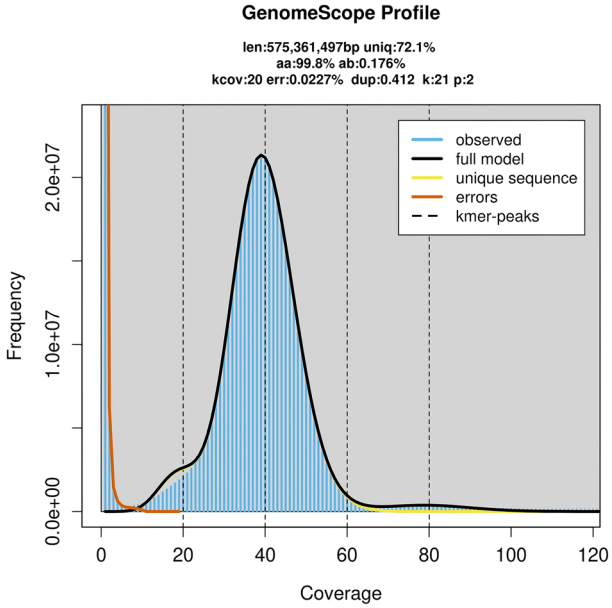


Fig. 2 21-mer frequency distribution of the *Troglonectes daqikongensis* genome.

Assembly level	Total length (bp)	N50 (bp)	N90 (bp)
Contig	566947446	8092	1993
Scaffold	567074799	8225	2015

Table 2. Summary statistics of the *Troglonectes daqikongensis* draft genome assembled from short reads.

complete circular mitochondrial genome assembly of 16,579 bp in length, which contained the full complement of 37 mitochondrial genes, comprising 13 protein-coding genes, 22 tRNAs, and 2 rRNAs (Fig. 3). Comparative sequence analysis demonstrated perfect concordance between the assemblies derived from HiFi long-reads and NGS short-reads in coding regions, thereby mutually validating the accuracy of both assembly methods.

Chromosome-level Genome Assembly. *De novo* genome assembly was performed by integrating PacBio HiFi long-read sequencing data with Hi-C chromatin conformation information. Raw Hi-C reads were quality-filtered using fastp (v0.23.4). PacBio HiFi reads were assembled into contigs using hifiasm (v0.25.0-r726; parameters: --h1/–h2)[29]. Following the initial assembly, potential contaminant sequences were identified and removed using FCS-GX (v0.5.4; parameters: --tax-id 1886673)[30], which revealed that 0.4% of the initial assembly (by base count) constituted contaminant sequences. Redundant haplotypes were further collapsed using purge_dups (v1.2.5; parameters: -2 -T cutoffs -c PB.base.cov)[31], resulting in a final polished genome assembly of 615.17 Mb, comprising 101 contigs with a contig N50 of 20.92 Mb (Table 3).

Chromosome-level scaffolding was achieved using Hi-C data. Clean Hi-C reads, preprocessed with fastp (v0.23.4; parameters: -f 15), were aligned to the primary assembly using BWA-MEM (v0.7.17-r1188; parameters: -5SP -t 28)[32], and alignment files were processed with SAMtools (v1.17; parameters: -F 3340)[33] for sorting and filtering. The HapHiC pipeline (v1.0.6; parameters: --quick_view --correct_nrounds 2 --processes 8)[34,35] was employed to anchor, order, and orient contigs, producing a scaffolded assembly and a corresponding Hi-C contact map. Manual curation was performed using Juicebox (v2.17.00)[36], allowing real-time visualization and correction of scaffold orientations and connections. The refined assembly was then re-scaffolded using the HapHiC pipeline, with pseudo-chromosomes arranged in descending order of length. Hi-C heatmaps were generated using the plot module in HapHiC to visualize chromatin interaction patterns (Fig. 4).

The final chromosome-level assembly of *Troglonectes daqikongensis* spans 615.17 Mb, comprising 47 scaffolds with a scaffold N50 of 23.57 Mb (Table 3), and 97.96% of the assembled sequences were anchored onto 25 pseudo-chromosomes (Table 4, Fig. 4). This speculation is based on Hi-C interaction patterns, synteny analysis with closely related species, and the relatively conserved chromosome number within the family Nemacheilidae[37]. Nevertheless, 22 scaffolds remain unanchored to the chromosomes (Table 5).

Repeat annotation. The repetitive sequences in the *Troglonectes daqikongensis* genome were annotated using an integrated approach that combined both homology-based and *de novo* prediction methods.

For homology-based annotation, RepeatMasker (v4.1.5; parameters: -a -e rmblast -gff -s -species Cypriniformes)[38] was employed in hardmask mode, using the Repbase (http://www.girinst.org/repbase)[39] and Dfam (https://www.dfam.org)[40] databases as reference libraries.

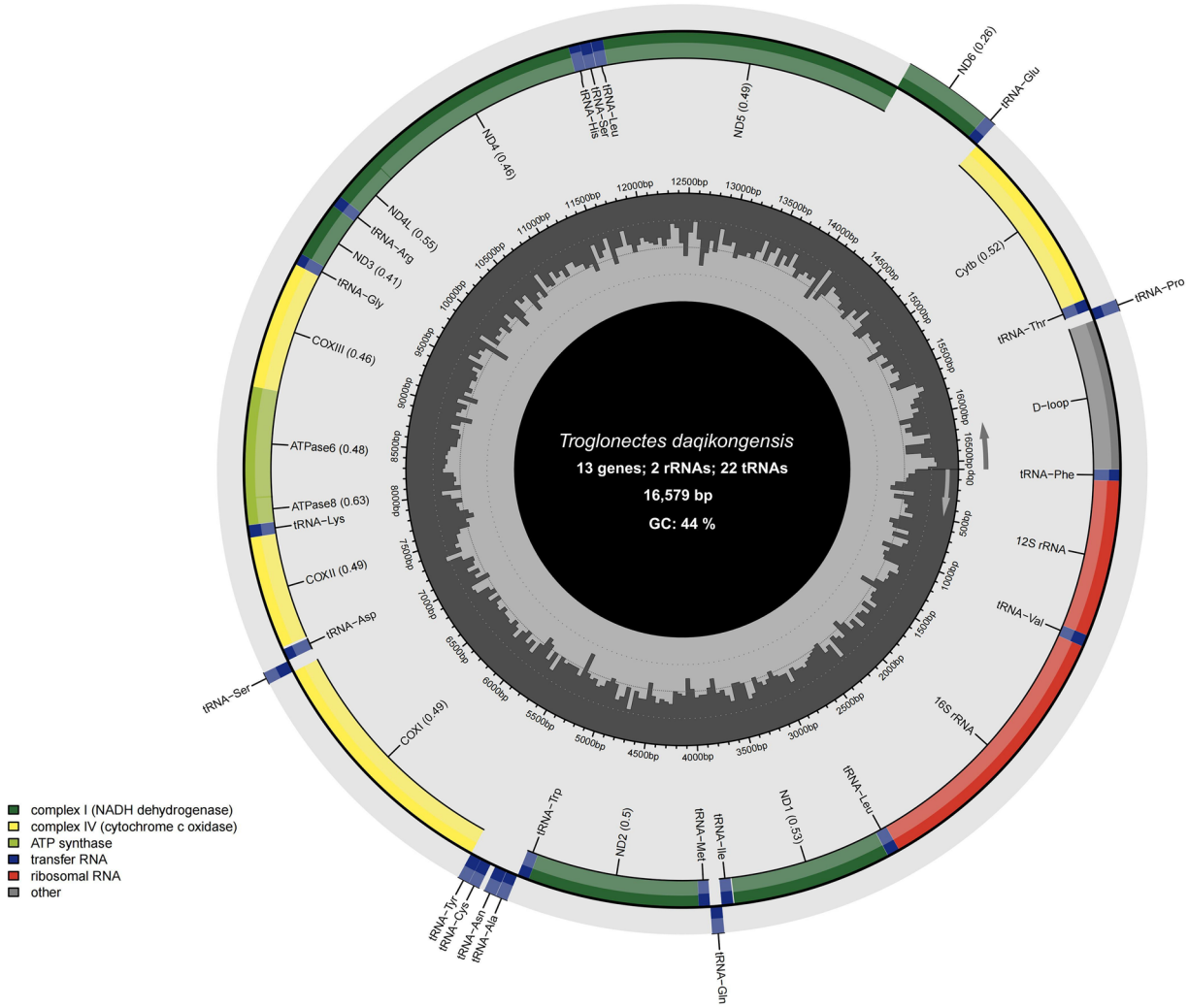


Fig. 3 Circular mitochondrial genome map of *Troglonectes daqikongensis*, generated from PacBio HiFi reads.

Statistic	Hifiasm assembly	FCS-GX Filtering	Purge_dups	Hi-C scaffolding
Total length(bp)	654623112	651862613	615165660	615165660
Number of contigs	279	239	101	—
Contig N50 (bp)	20132969	20132969	20922685	—
Contig N90 (bp)	2994343	3046456	4844381	—
Contig L50	13	13	12	—
Contig L90	39	38	30	—
Number of scaffolds	—	—	—	47
Scaffold N50 (bp)	—	—	—	23570105
Scaffold N90 (bp)	—	—	—	20210658
Scaffold L50	—	—	—	12
Scaffold L90	—	—	—	23
GC Content (%)	39.03	39.01	39.04	39.04

Table 3. Assembly statistics of the *Troglonectes daqikongensis* genome at different processing stages.

For *de novo* prediction, a species-specific repeat library was first constructed. EDTA (v2.0.1; parameters:–species others–sensitive 1–anno 1–step all)[41] was used to identify long terminal repeat (LTR) elements and generate a high-quality, non-redundant LTR library. Non-LTR repetitive elements were identified using RepeatModeler (v2.0.4; parameters: -database DQKDQ_2515)[42], which generated a complementary repeat library. The two libraries were merged and de-redundantized using CD-HIT (v4.8.1; parameters: -c 0.9 -n 8)[43] to produce a final *de novo* repeat library.

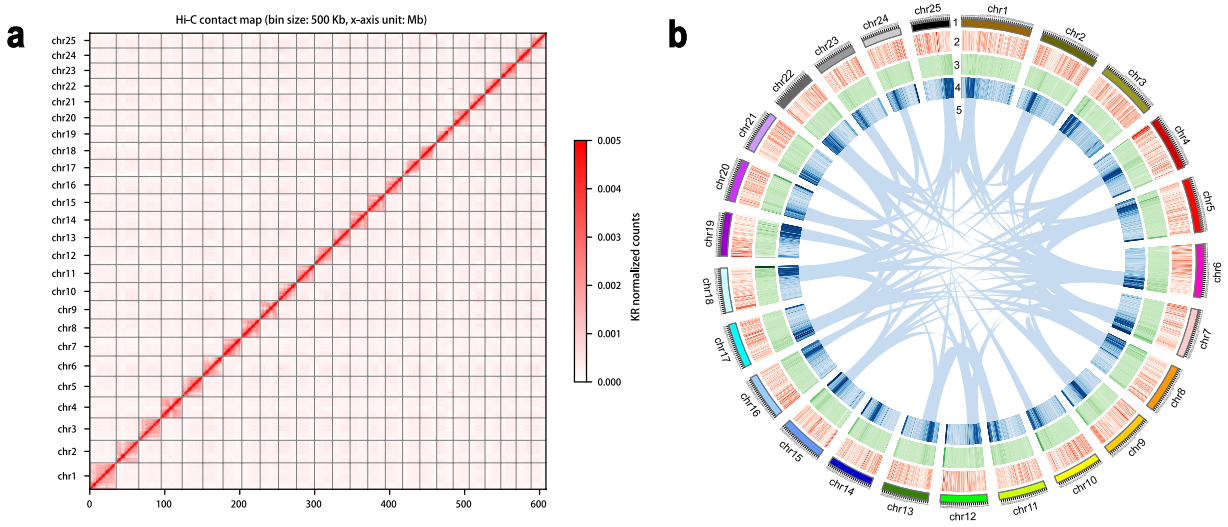


Fig. 4 Genome assembly overview of *Troglonectes daqikongensis*. (**a**) Hi-C contact heatmap showing chromosomal interactions, indicating chromosome-level assembly. (**b**) Circos plot illustrating genomic features from outermost to innermost layers: (1) chromosome layout; (2) heatmap of gene density; (3) GC content distribution; (4) repeat content distribution; (5) syntenic blocks indicating intra-genomic collinearity.

Chr ID	GC Content (%)	Length (bp)	Proportion of Genome (%)
chr1	38.95	35080718	5.7
chr2	38.69	29839626	4.85
chr3	38.87	29596051	4.81
chr4	38.79	27840538	4.53
chr5	39.1	27390433	4.45
chr6	39.1	26741620	4.35
chr7	39.59	24802226	4.03
chr8	39.57	24318037	3.95
chr9	38.91	24247197	3.94
chr10	39.14	23857597	3.88
chr11	38.96	23856846	3.88
chr12	38.95	23570105	3.83
chr13	39	23490636	3.82
chr14	39	23409585	3.81
chr15	39.11	23027567	3.74
chr16	39.11	22933898	3.73
chr17	39.16	22571587	3.67
chr18	39.33	22416409	3.64
chr19	38.94	21934814	3.57
chr20	39.26	21301009	3.46
chr21	39.2	20970236	3.41
chr22	39.28	20543987	3.34
chr23	39.09	20210658	3.29
chr24	39.11	19516929	3.17
chr25	39.38	19143086	3.11
total	39.09	602611395	97.96

Table 4. Chromosome-level assembly statistics of the 25 Pseudo-chromosomes of *Troglonectes daqikongensis*.

The genome, previously hardmasked based on the homology-based annotation, was then re-annotated using RepeatMasker (v4.1.5; parameters: -a -e rmblast -s -lib combined_TE.clean.fa -html -gff) with the *de novo* repeat library, followed by a second round of hardmasking. By integrating results from both annotation strategies, a total of 276.51 Mb of repetitive sequences were identified, accounting for 44.95% of the assembled genome (Table 6). Finally, after filtering out simple repeats, the genome was softmasked using the maskFile.pl script from the RepeatMasker (v4.1.5) package to prepare for downstream gene structure annotation.

Scaffold ID	GC Content (%)	Length (bp)	Proportion of Genome (%)
scaffold_29	37.80	4844381	0.7875
scaffold_28	38.34	4266911	0.6936
scaffold_36	28.15	846708	0.1376
scaffold_27	37.82	635458	0.1033
scaffold_26	38.53	460543	0.0749
scaffold_35	28.15	454490	0.0739
scaffold_32	28.40	389688	0.0633
scaffold_38	41.66	342579	0.0557
scaffold_37	42.55	63600	0.0103
scaffold_31	25.01	56306	0.0092
scaffold_40	42.52	33468	0.0054
scaffold_41	42.43	31672	0.0051
scaffold_34	42.60	30559	0.0050
scaffold_33	27.94	29700	0.0048
scaffold_39	37.98	21623	0.0035
scaffold_30	43.69	16579	0.0027
scaffold_42	39.22	5000	0.0008
scaffold_43	41.76	5000	0.0008
scaffold_44	39.26	5000	0.0008
scaffold_45	39.52	5000	0.0008
scaffold_46	46.10	5000	0.0008
scaffold_47	40.12	5000	0.0008

Table 5. Summary statistics of the unanchored scaffolds in the *Troglonectes daqikongensis* genome assembly.

Repeat Type	Length(bp)	Percentage (%)
SINEs	2583296	0.42
LINEs	23241340	3.78
LTR elements	47582854	7.74
DNA transposons	155719142	25.31
Unclassified	30956519	5.03
Other	16426450	2.67
Total	276509601	44.95

Table 6. Summary of repetitive elements in the *Troglonectes daqikongensis* genome.

Gene prediction and functional annotation. Protein-coding genes in the *Troglonectes daqikongensis* genome were annotated by integrating *de novo* prediction, homology-based prediction, and transcriptome-based evidence.

For transcriptome-based gene prediction, total RNA was extracted from nine tissues, including the brain, gill, kidney, liver, muscle, ovary, skin, swim bladder, and olfactory organ. Raw RNA-seq reads were quality-filtered using fastp (v0.23.4; parameters: -f 15), and the resulting clean reads were aligned to the reference genome using HISAT2 (v2.2.1; parameters:–dta–new-summary)[44]. The alignment files were sorted and indexed with SAMtools (v1.17) to produce BAM files for downstream analysis. A genome-guided transcriptome assembly was then performed using Trinity (v2.9.1; parameters: --genome_guided_bam --genome_guided_max_intron 50000)[45], guided by the aligned reads. The resulting transcript assemblies were processed using PASA (v2.5.3; parameters: -C -R --ALIGNERS minimap2,blat,gmap)[46] with the alignAssembly workflow to refine transcript structures and alignments. Open reading frames (ORFs) were predicted using TransDecoder (v5.5.0)[47], and full-length coding sequences were extracted. The final set of transcriptome-derived gene models was generated in GFF3 format, representing predicted gene structures and complete ORFs.

For homology-based gene prediction, genome assemblies and corresponding GFF annotation files were retrieved from NCBI for six Cypriniformes species: *Danio rerio* (GCF_049306965.1), *Carassius auratus* (GCF_003368295.1), *Misgurnus anguillicaudatus* (GCF_027580225.2), *Triplophysa dalaica* (GCF_015846415.1), *Triplophysa rosa* (GCF_024868665.1), and *Megalobrama amblycephala* (GCF_018812025.1); and three additional species from other orders: *Astyanax mexicanus* (GCF_023375975.1; Characiformes), *Ictalurus punctatus* (GCF_001660625.3; Siluriformes), and *Oryzias latipes* (GCF_002234675.1; Beloniformes). For each species, the longest transcript isoform per gene was extracted and translated into protein sequences. These protein sequences were aligned to the *Troglonectes daqikongensis* genome using tblastn (v2.14.1; parameters: -evalue 1e-5 -max_target_seqs. 5 -outfmt 6)[48] to identify candidate genomic loci. The resulting alignments were further refined

Feature	Value
Number of protein-coding genes	24652
Mean gene length (bp)	22783
Mean exon length (bp)	253
Mean CDS length (bp)	1711
Mean intron length (bp)	2340

Table 7. Summary statistics of gene annotation for the *Troglonectes daqikongensis* genome.

Annotation Source	Number of Annotated Genes	Percentage of Total Genes (%)
eggNOG (emapper)	23793	96.52
Interpro	24434	99.12
KEGG	17542	71.16
NR	24529	99.5
Swiss-Prot	20795	84.35
TrEMBL	24496	99.37
Total annotated	24625	99.89

Table 8. Summary of functional annotation results for the *Troglonectes daqikongensis* genome.

using GeneWise (v2.4.1; parameters: -nosplice_gtag -gff)[49] to accurately predict gene structures, which were output as GFF files representing homologous gene models.

For *de novo* gene prediction, we employed the integrated pipeline BRAKER3 (v3.0.6; parameters:–soft-masking)[50]. This pipeline combines the self-training algorithm GeneMark-ES and the homology-based predictor AUGUSTUS. The AUGUSTUS algorithm was trained using two lines of evidence: (1) transcriptomic data aligned to the genome with HISAT2 (v2.2.1; parameters:–dta) and (2) protein sequences from closely related species as homologous evidence. Based on the species-specific parameters generated from this training, AUGUSTUS was subsequently applied to the genome to predict protein-coding gene structures *de novo*. This process generated two primary outputs: (1) the raw BRAKER consensus output (braker.gtf), which was post-processed with AGAT (v1.4.1; https://github.com/NBISweden/AGAT) to produce a high-confidence, non-redundant annotation set; and (2) the AUGUSTUS-specific output (augustus.hints.gtf), which was converted to GFF3 format and incorporated as a complementary candidate gene set for evidence integration in the final annotation.

Finally, all gene prediction evidence from transcriptome-based, homology-based, and *de novo* approaches was integrated using EVidenceModeler (EVM, v2.1.0)[51]. The EVM-generated gene models were filtered based on the completeness of the open reading frame (ORF) and a minimum protein length of 50 amino acids. These high-confidence candidates were then validated and refined using PASA (v2.5.3; parameters: -A -t Trinity-GG_modfiy.fasta) to (i) identify and exclude models lacking transcript support based on real RNA-Seq evidence, and (ii) incorporate untranslated regions (UTRs) and alternative splicing isoforms. This integrative annotation strategy resulted in the identification of 24,652 protein-coding genes (Table 7).

Functional annotation of the predicted protein-coding genes was performed by aligning the protein sequences against the NR[52] and UniProtKB[53] (Swiss-Prot and TrEMBL) databases using DIAMOND (v2.1.8.162; parameters: --evalue 1e-5 --max-target-seqs. 1 --outfmt 6)[54]. Gene motifs and domains were identified with InterProScan (v5.72-103.0; parameters: -dp -goterms -b output_prefix)[55]. Additional functional annotation was conducted using eggNOG-mapper (v2.1.12; parameters: -m mmseqs --evalue 1e-5)[56] based on the eggNOG database (v5.0.2)[57]. To further enhance the functional annotation, the KEGG online tool GhostKOALA (https://www.kegg.jp/ghostkoala/)[58] was also used to perform KEGG Orthology (KO) assignments based on the KEGG database[59]. As a result, 24625 protein-coding genes were functionally annotated, representing 99.89% of all predicted genes (Table 8, Fig. 5).

Non-coding RNA Annotation. Non-coding RNAs (ncRNAs) were annotated using a combination of model-based prediction and database-driven approaches. Ribosomal RNAs (rRNAs) were identified using Barrnap (v0.9; https://github.com/tseemann/barrnap; parameters: --kingdom euk --outseq rRNA.fasta) with the eukaryotic prediction model, while transfer RNAs (tRNAs) were annotated with tRNAscan-SE (v2.0.12; parameters: -E --gff)[60]. To detect other classes of ncRNAs, the Rfam database was employed in conjunction with cmscan (v1.1.4; parameters: --cut_ga –rfam --nohmmonly)[61] to identify microRNAs (miRNAs), small nuclear RNAs (snRNAs), and other non-coding elements. In total, 4,774 rRNA genes, 9,759 tRNA genes, 896 miRNA genes, and 1,652 snRNA genes were identified (Table 9).

Data Records

All raw sequencing data, including PacBio HiFi, Hi-C, RNA-seq, and short-read NGS reads, have been deposited in the NCBI Sequence Read Archive (SRA) under BioProject accession PRJNA1284592 (https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1284592) and the umbrella SRA Study accession SRP597166[62]. The final genome assembly has been deposited in the NCBI GenBank database under the accession number GCA_052426585.1[63].

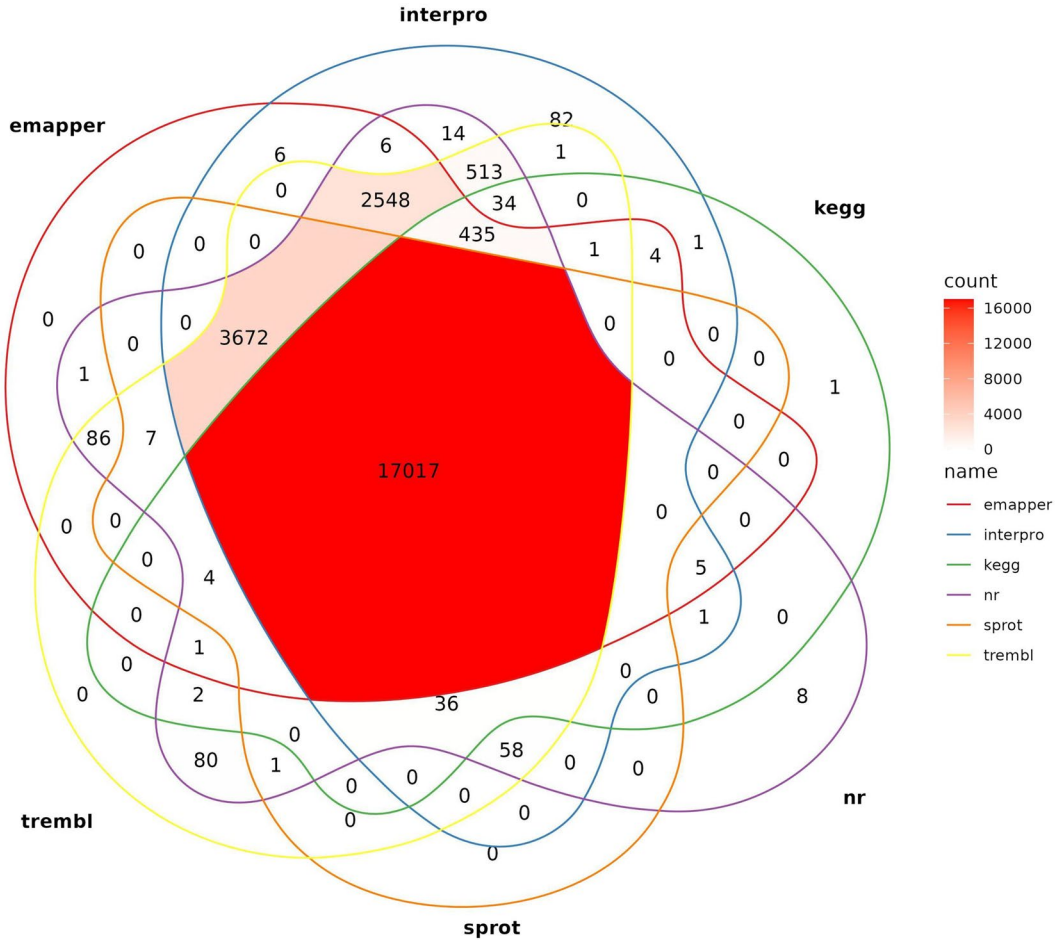


Fig. 5 Venn diagram of functional annotation of predicted protein-coding genes in the *Troglonectes daqikongensis* genome based on six databases.

RNA Type	Subtype	Copy number	Total length(bp)	Mean length(bp)
rRNA	18S_rRNA	35	43445	1241.29
	28S_rRNA	13	35712	2747.08
	5.8S_rRNA	6	918	153
	5S_rRNA	4720	531794	112.668
tRNA	—	9759	725116	74.3023
miRNA	—	896	65290	72.87
snRNA	splicing	1390	209355	150.62
	CD-box	168	21626	128.73
	HACA-box	76	11329	149.07
	scaRNA	11	2163	196.64
	Other	7	380	54.29

Table 9. Summary of non-coding RNAs in the *Troglonectes daqikongensis* genome.

The mitochondrial genome assembled from MGI short-read NGS data has been deposited in GenBank under accession number PV866694[64]. The mitochondrial genome assembled from PacBio HiFi long-read data has been deposited under accession number PX308854[65]. The genome annotation files, including repeat mask, structural gene models, functional assignments, non-coding RNA predictions, and an optional Braker-derived annotation set (braker.gff3), are publicly available via the Figshare repository[66].

Data Overview

This study assembled a chromosome-level genome for the obligate cave loach *Troglonectes daqikongensis*. The primary data and assembly outcomes are summarized as follows:

Data Type	Mapping Rate (%)	Genome Coverage (%)	Genome Coverage >10 ×(%)
Short-read NGS	99.65	100	99.91
Pacbio HiFi	99.91	100	99.95

Table 10. Mapping statistics of short-read and long-read sequencing data to the *Troglonectes daqikongensis* genome.

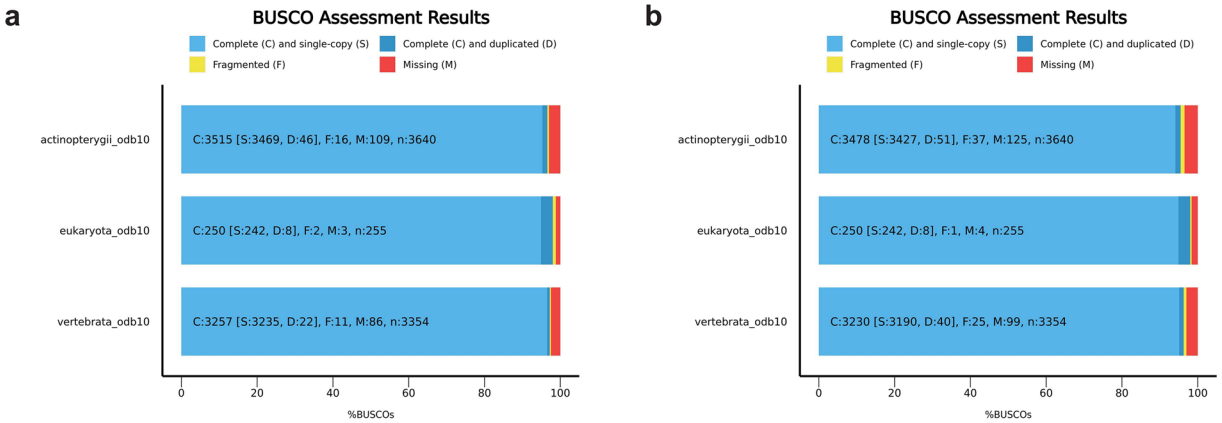


Fig. 6 Assessment of genome assembly and annotation completeness using BUSCO. (**a**) BUSCO evaluation results for the *de novo* genome assembly. (**b**) BUSCO evaluation results for the structural annotation of protein-coding genes.

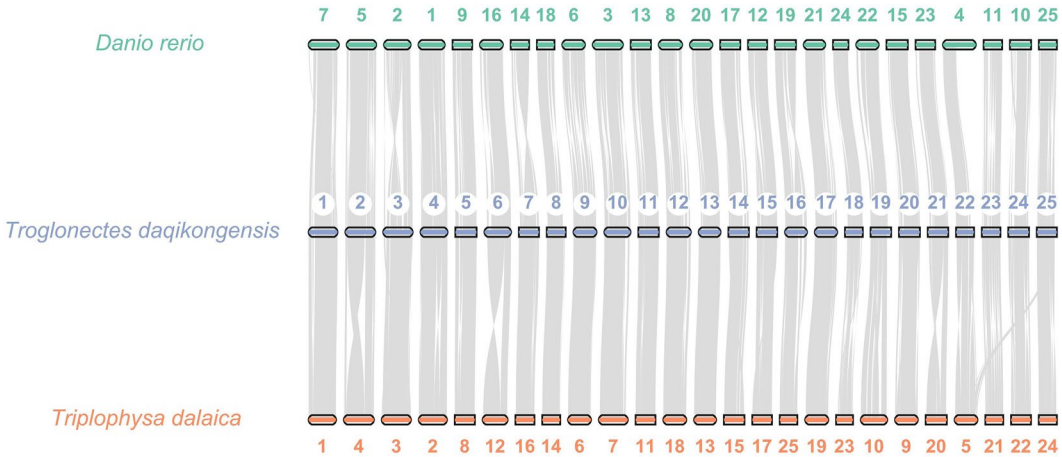


Fig. 7 Synteny relationships among *Troglonectes daqikongensis*, *Danio rerio*, and *Triplophysa dalaica*.

1. Raw Sequencing Data: A total of 41.17 Gb of PacBio HiFi long-read data, 63.95 Gb of Hi-C chromatin interaction data, 32.03 Gb of MGI short-read NGS data, and 73.77 Gb of RNA-seq data from nine distinct tissues (MGI platform) were generated.
2. Genome Assembly: The final genome assembly was 615.17 Mb in size, comprising 47 scaffolds with a contig N50 of 20.92 Mb and a scaffold N50 of 23.57 Mb. The assembly completeness, as assessed by BUSCO (actinopterygii_odb10), reached 96.6%. Additionally, a complete circular mitochondrial genome of 16,579 bp was assembled.
3. Genome Annotation: Repetitive Sequences: Analysis revealed that 276.51 Mb (44.95%) of the genome consists of repetitive elements. Protein-Coding Genes: A total of 24,625 protein-coding genes were predicted through a structural annotation pipeline. BUSCO evaluation (actinopterygii_odb10) indicated a high annotation completeness of 95.5%. Functional Annotation: Functional assignments were successfully made for 99.89% of the predicted genes using a combination of multiple databases. Non-coding RNAs: The annotation identified 4774 rRNAs, 9759 tRNAs, 896 miRNAs and 1652 snRNAs.

Technical Validation

To assess the quality of the *Troglonectes daqikongensis* genome assembly, short-read sequencing data were aligned to the assembled genome using BWA (v0.7.17-r1188; parameters: -R "@RG\tID:mgi\tSM:sample"). A total of 99.65% of reads successfully mapped to the genome, achieving 100.00% coverage, with 99.91% of the genome covered at a depth greater than 10×. In parallel, PacBio HiFi long reads were aligned using Minimap2 (v2.26-r1175; parameters: -ax map-hifi)[67], resulting in a mapping rate of 99.91%, full genome coverage (100.00%), and 99.95% of the genome covered at >10 × depth (Table 10).

In addition, the completeness of the assembled genome was assessed using BUSCO (v5.5.0)[68]. Using the eukaryota_odb10, vertebrata_odb10, and actinopterygii_odb10 lineage datasets, 98.0%, 97.2%, and 96.6% of complete BUSCOs were identified, respectively, supporting the high quality of the genome assembly. The completeness of the gene annotation was also evaluated with BUSCO (v5.5.0), which identified 98.0%, 96.3%, and 95.5% complete BUSCOs in the same three datasets, demonstrating the robustness and accuracy of the annotation results (Fig. 6).

To further assess the accuracy of the genome assembly and annotation, and to investigate genome structure conservation, synteny analysis was conducted using JCVI (v1.1.12)[69] among *Troglonectes daqikongensis*, *Danio rerio* (GCF_049306965.1), and *Triplophysa dalaica* (GCF_015846415.1). The results revealed a high degree of gene synteny between *Troglonectes daqikongensis* and both *Danio rerio* and *Triplophysa dalaica*, providing additional evidence for the completeness and accuracy of the assembled and annotated genome (Fig. 7).

Code availability

No custom code or scripts were used in this study. All analyses were conducted using publicly available software, with versions and tool names detailed in the Methods section. All commands and analytical procedures strictly followed the official user manuals and standard protocols provided by the respective software. For tools where specific parameters were not explicitly stated, default settings recommended by the developers were used.

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Author contributions

Guohuan Su: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing; Jun Xu: Supervision; Houhua Bing: Conceptualization, Investigation, Formal analysis, Methodology, Data curation, Writing – original draft, Visualization; Wenyu Fang: Investigation, Data curation; Jiayang He: Investigation; Zichen Xiong: Investigation; Yongrui Lu: Formal analysis; All authors: Writing – review & editing, Approval of final version. Houhua Bing is recognized as the first author. Guohuan Su is the corresponding author.

Competing interests

The authors declare no competing interests.

Additional information

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